Review retrospective: *the moment-ladder base* — iteratedDeriv $k\,Z = G_k$

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Abstract

A soundness review of `Laplace/OneD/AnharmonicPartitionDerivGeneralH.lean`: the file that defines the weighted partition family $G_n(h) = \int(-(tx))^n e^{-t(L+hx)}$ and proves the moment-ladder identity iteratedDeriv $k\,(G_0) = G_k$ on $|h| < 1$, with the $Z''(0), Z'''(0)$ corollaries. Result: **a clean fidelity pass with no edits.** The general-h dominated-differentiation step and the induction are correct, and both `open`s are load-bearing.

1 Setting

This is the keystone of the anharmonic cumulant machinery: it upgrades the $h = 0$ derivative step of AnharmonicPartitionHigherDeriv to a general-h statement, then promotes the single derivative to the full iterated-derivative identity iteratedDeriv $k\,Z = G_k$ — which is what makes the cumulant files (AnharmonicFourthCumulant) genuinely about derivatives of Z, not just of integral expressions that happen to coincide.

2 What was reviewed

The definition `weightedPartition` and eight declarations: the bare-exponential domination, integrand integrability, `weightedPartition_hasDerivAt` ($G_n' = G_{n+1}$ at general $|h| < 1$), `weightedPartition_deriv`, the induction `iteratedDeriv_weightedPartition_zero` (iteratedDeriv $k\,(G_0) = G_k$), and the $Z''(0) = \int(tx)^2 e^{-tL}$, $Z'''(0) = -\int(tx)^3 e^{-tL}$ corollaries.

3 Fidelity / soundness findings

Sound; checked end to end. The definition is G_n. The general-h step is the right neighbourhood upgrade of the $h = 0$ argument: coercivity data, `ball 0 1`$\in \mathcal{N}\, h_0$, AE-measurability of the family and the derivative, an integrable majorant, the uniform derivative bound on the ball, pointwise `HasDerivAt`, and `hasDerivAt_integral_of_dominated_loc_of_deriv_le`. The induction is the standard germ-locality move: $\text{iteratedDeriv}(k+1)(G_0) = \text{deriv}(\text{iteratedDeriv}\,k\,(G_0)) = \text{deriv}(G_k) = G_{k+1}$, the middle step via $=_{\mathrm{f}}$ $[\mathcal{N} h]$ from the induction hypothesis on the ball. $Z''(0) = G_2(0) = \int(tx)^2$, $Z'''(0) = G_3(0) = -\int(tx)^3$. The reviewer confirmed all five points. Crucially, because the file proves iteratedDeriv $k\,Z = G_k$, the corollary titles "second/third h-derivative of Z" are fully warranted here — closing the report-only caveat noted on the $h = 0$ sibling.

4 Simplifications applied

None — the bucket was empty. `open MeasureTheory` (bare $\int$ and measure API) and `open Topology` (bare $\mathcal{N}$ and $=_{\mathfrak{n}}$) are both load-bearing; there is no `open Real`; the single import is used; and the dense `weightedPartition_hasDerivAt` proof has no dead local or no-op step. No worktree.

5 Disagreements and open questions

None.

6 Lessons

- **Proving the iterated-derivative identity is what licenses the "$Z^{(k)}$" prose.** The $h = 0$ sibling proved derivatives of integral *expressions* and had to hedge its "Z''" wording; this file proves iteratedDeriv $k\, Z = G_k$ outright, so the same wording is exact. When a docstring says "the k-th derivative of Z", the licensing fact is precisely an in-scope `iteratedDeriv` identity — check for it before flagging the prose either way.
- **Two integral-side files in a row, both `MeasureTheory`-load-bearing.** The measure-vs-algebra split continues to hold per-file: the differentiation-under-the-integral base and this general-h/iterated promotion both carry bare $\int$, in contrast to the pure cumulant algebra above them.

7 Follow-ups

- The remaining anharmonic arc (`AnharmonicGibbsRegularity`, `AnharmonicGibbsObservableMonomials`, `AnharmonicKappa3Affine`) and the harmonic regularity/observable/perturbed-moment files.